

ME2 DMF Autumn Project Report - Group 35

Yasmine Boury

Jack Broster

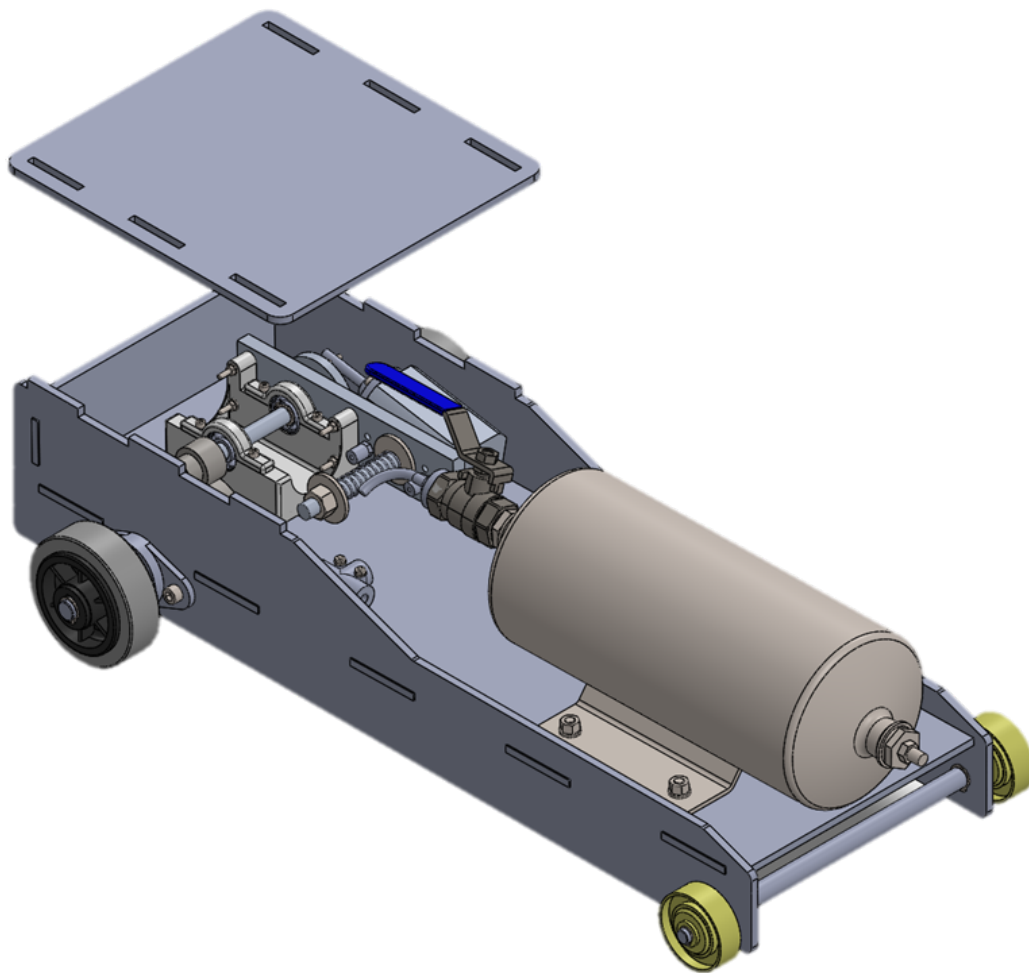
Arnav Goel

Max Hurlow

Jaime Lopez Ruiz

Department of Mechanical Engineering, Imperial College London

December 2023



1 Executive Summary

This report details the design, manufacturing and assembly of a compressed air-powered toy car aiming to navigate a 25-meter course in the shortest time while optimising quality, cost-effectiveness and ease of use. Our team was given basic materials, the possibility of acquiring components from a supplier and access to the Student Teaching Workshop, as well as the freedom of choice in terms of our procedure for the 10 weeks the project lasted.

Following the development of a PDS, which allowed for an initial organised collection of ideas and guidelines, our team engaged in several days of brainstorming where we presented diverse approaches to solve the problem, considering the advantages and disadvantages of each.

Ultimately, we selected an oscillating compressed air engine, an interesting and perhaps unconventional solution which tested our ability in both the workshop and during the design process whilst providing invaluable insight into the workings of a piston mechanism from the perspective of the manufacturer.

Contents

1	Executive Summary	ii
2	Introduction	1
3	Design Development	1
3.1	PDS and Concept Development	1
3.2	Chassis Design and Stress Analysis	3
4	Functional and Operational Description	5
5	Performance Validation	7
5.1	Gear Selection	7
5.2	Axle Stress Analysis	8
5.3	Bearing Selection	9
5.4	Flywheel	9
6	Assembly Instructions	10
7	Conclusion	11
	References	11
A	Measurement Report	12
A.1	Cylinder	12
A.2	Distributor	13
B	Project Timeline	15
C	Engineering Drawings	15
C.1	Rear Axle	15
C.2	General Assembly	15
C.3	Engine Subassembly	18

2 Introduction

Faced with the challenge of coming up with a toy car that complies with the restrictions detailed in the PDS, we opted for a piston mechanism. Specifically, we chose one of 2 power strokes per cycle known as the wobbler engine due to its characteristic ‘wobbling’, or rocking motion during operation, and transmitted the drive to the back wheels using gears.

The rest of this report aims to document the whole process, including reasoning and justification for the decisions made, from initial concepts to final product. The roles assigned to our five team members are as follows: Jack Broster, project manager and chassis design lead; Yasmine Boury, procurement and bearing calculations; Jaime Lopez Ruiz, drivetrain lead; Arnav Goel, flywheel and engine output; and Max Hurlow, head of engine design and CAD.

3 Design Development

3.1 PDS and Concept Development

The project brief was written up into a formal PDS to provide measurable goals that the final product must meet. This is provided in Table 1. Some PDS items are taken from the project brief (Mohammed 2023).

Following completion of the PDS, multiple concepts were presented, falling into three distinct categories: turbines, reciprocating engines and jet propulsion. Each of these categories had their own distinct merits and drawbacks which required careful comparison as detailed below.

The simple jet propulsion system would

have been the easiest to manufacture in the Student Teaching Workshop (STW) with a very low number of complex parts to machine. However, after experimental review of the flow rate from the tank, this concept was quickly discarded. The measured discharge time of the opened tank was around one second, far less than sufficient to travel the 25 m distance required by the PDS.

The turbine would be a significant improvement on the jet propulsion concept, providing a much more effective use of the available air by transmitting the power through a drivetrain and scaling the output torque. However, challenges began to appear when considering the airflow through a turbine. This is a complex fluid dynamics problem involving compressibility and turbulence, which was beyond the scope of our knowledge at this point in the year. As a result, the piston engine was introduced as an alternative solution. With some prior knowledge of existing designs and their operation, this presented a much stronger basis for calculations.

However, the reciprocating engine concept is not without its drawbacks, the most significant being the increased weight from the higher number of components and the need for many to be manufactured from steel or aluminium. The turbine and housing could be manufactured from ABS plastic using a 3D printer and would therefore be lighter. However, the airtightness achieved by 3D printing would be of a lower quality than metal components meaning a higher volume of air would be wasted.

Table 1: *The PDS the project was based on*

Element	Criteria	Verified by
Design Requirements		
Needs	To provide a small toy vehicle powered by a compressed air tank which can easily be filled and operated at a gauge pressure of 5.5 bar.	n/a
Environment	To run along a smooth floor (CAGB concourse), may be exposed to small debris.	n/a
Performance	Must travel a 20-25 m straight track in the fastest possible time. No steering required.	Testing
Size	Cannot exceed 500 mm height or 400 mm width. No maximum length.	CAD model
Weight	Minimised to reduce torque requirement and energy losses.	CAD model
Design	Must use all the 4 wheels, air storage container and release valve provided.	Design review
	Transmission elements must be completely covered, so no moving parts are exposed.	CAD model
	Release valve and pneumatic connection must be accessible without removing any other parts such that the car can be operated and refilled easily	Design review
Aesthetics	Aesthetically pleasing body design to make the product attractive.	Customer survey
Restrictions	Design must contain at least: • 1 manually turned part • 1 manually milled part • 1 laser cut part CNC machining can be used for two units of one part.	Design review
Life		
Service life	The toy must perform for a minimum of 100 hours.	Calculation/testing
	Must be able to withstand an impact at its maximum speed without damage.	Testing
Procurement and Manufacture		
Purchased parts	All purchased parts must come from RS components (https://uk.rsonline.com/).	Design review
	Cost and material waste should be minimised	Design review
Manufacture	Must use readily available manufacture processes (turning, milling, laser cutting, 3D printing).	Design review

Following some preliminary research into the sealing and finishes required to make a functional airtight engine, we concluded that it would be possible to machine the engine parts to an appropriate tolerance in the STW. However, this would significantly increase the manufacture time of each part, meaning careful planning would be critical if this concept were to be chosen.

On consideration of the merits of the turbine and reciprocating engine concepts it was decided that priority should be given to the efficiency of air use and the reliability of sealing to the detriment of increased weight and part complexity. This meant that an engine design was selected and the task of reducing weight was assigned to the design of the chassis.

With the goal of reducing part count and complexity, two engine designs presented themselves where the intake and exhaust could be controlled by the motion of the piston-cylinder arrangement itself, removing the need for external timing elements. The first of these concepts is known as an oscillating, or ‘wobbler’ engine where the cylinder is centrally pivoted and oscillates to appropriately align to the intake and exhaust ports. The detailed operation of this engine will be described in the next section.

The second option functions using a ball bearing and seal as an inlet valve as shown in Figure 1. As the piston reaches the top of its travel, it pushes the ball bearing up and allows air to flow into the cylinder for the power stroke. The ball bearing then falls under the action of gravity, resealing the inlet as the piston travels to the bottom of its stroke where it passes another hole that allows the air to exhaust. It would then travel back to the top of its motion with inertia to

restart the cycle.

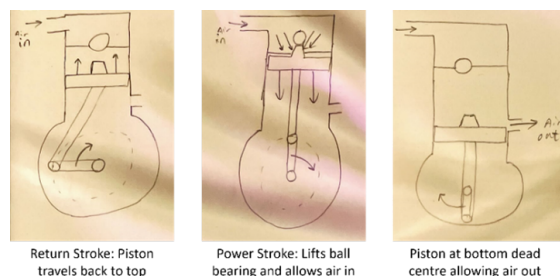


Figure 1: The basic operation of the ball bearing engine

This option had some disadvantages when compared to the ‘wobbler’ engine, the most prominent being the far shorter and less frequent power strokes introducing more opportunity for stalling if sufficient kinetic energy was not available. Also, as gravity causes the ball bearing to return to the closed position, the possible engine orientations are restricted, imposing further geometric constraints on the chassis.

The ‘wobbler’ design was chosen as our final concept because it makes the most efficient use of the available air with minimal leakage and has the potential to be configured as double acting to provide two power strokes per revolution.

3.2 Chassis Design and Stress Analysis

To compensate for the heavier drive mechanism, it was imperative to reduce the weight of the chassis. With this in mind, the original idea to have a frame of metal bars on which the components would be mounted developed into a much lighter acrylic chassis where the components would be mounted on a baseplate supported with side walls and

covered by a lid. This acrylic sheet was easily laser cut which saved a significant amount of manufacturing time, enabling the precision of the engine parts to be maximised. However, it was necessary to verify that the acrylic would be sufficient to support the heavy components by calculating the bending stresses within the material as detailed here.

The 201×5×637 mm baseplate was considered from both the side and front views independently and in each was modelled as a simply supported beam to find the maximum stress was experienced along the 637 mm length of the vehicle. The largest stress is caused by a moment of -1.9 Nm as read from the bending moment diagram shown in Figure 2. This moment causes a stress of 2.23 MPa in the bottom surface as calculated from Equation 1 which results in a safety factor of 31 when compared to the maximum permissible tensile stress of 70 MPa (Engineering Toolbox 2003).

$$\sigma = \frac{My}{I} \quad (1)$$

The chassis originally placed the shafts above the baseplate so that the gears would be enclosed in a housing with the other moving parts. However, after reviewing the developing CAD model, it was discovered that this did not give enough clearance for the front wheels to touch the floor as the side view illustrates in Figure 3. Note the 10 mm dimensions are fixed so there is sufficient material between any two holes.

To solve this problem, the axles were moved below the baseplate with a cut out for the final gear to sit between the engine driveshaft and the rear axle. The baseplate was also inclined at a 1° angle to ensure the

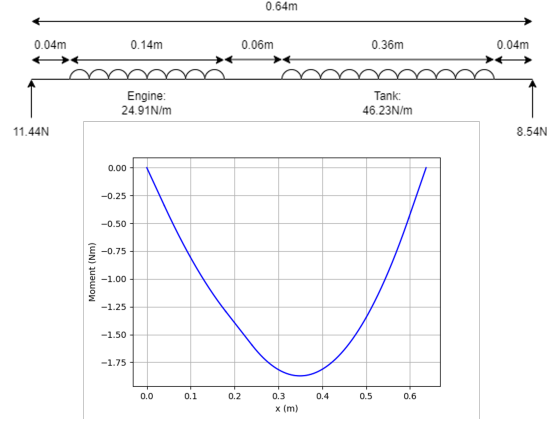


Figure 2: The beam representation used in the calculations, and the corresponding bending moment diagram

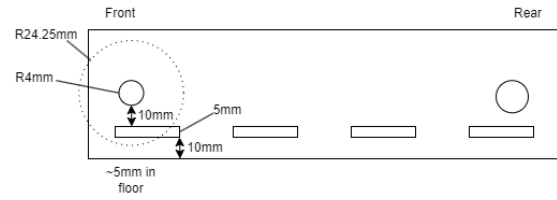


Figure 3: Before - the front wheels cannot touch the ground

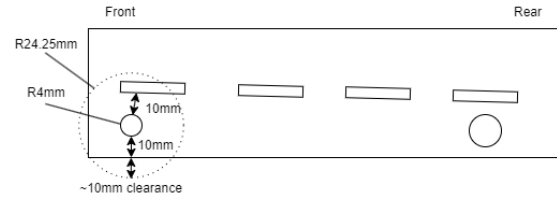


Figure 4: After - the baseplate has been moved to allow clearance between drivetrain elements and the floor

10 mm spacing could be preserved. The final side profile is shown in Figure 4.

4 Functional and Operational Description

The power source for the vehicle is a double-acting oscillating cylinder engine, sometimes known as a ‘wobbler’. This engine design was used to power several ships in the 19th century, most notably the first iron-hulled steam ship, the *Aaron Manby*. The engine used for this project, while significantly smaller and with a different working fluid, runs on the same principles as these full-sized engines.

The cylinder, which has two holes at its top and bottom, has a flat side, which mates with a flat distributor plate, as shown in Figure 5. The distributor has two holes, one for each hole in the cylinder’s flat face. One of these holes is an inlet port, the other an exhaust. The cylinder is fixed to the distributor by a central pivot.

A graphical depiction of a full rotation is shown in Figure 6. We will start our explanation with the crankshaft at the 3 o’clock position. In this position, the port at the top of the cylinder aligns with the inlet port of the distributor. This allows compressed air to flow into the cylinder, depressing the piston and rotating the crankshaft clockwise. The movement of the piston also causes the cylinder to rotate. When the crankshaft reaches bottom dead centre, the port on the cylinder no longer aligns with the inlet, and no more air can enter. As the crankshaft continues to rotate, the cylinder port aligns with the other port on the distributor, the exhaust. This allows the air in the cylinder to escape as the piston starts to move upwards. After the crankshaft rotates past top dead centre, the inlet port is again opened

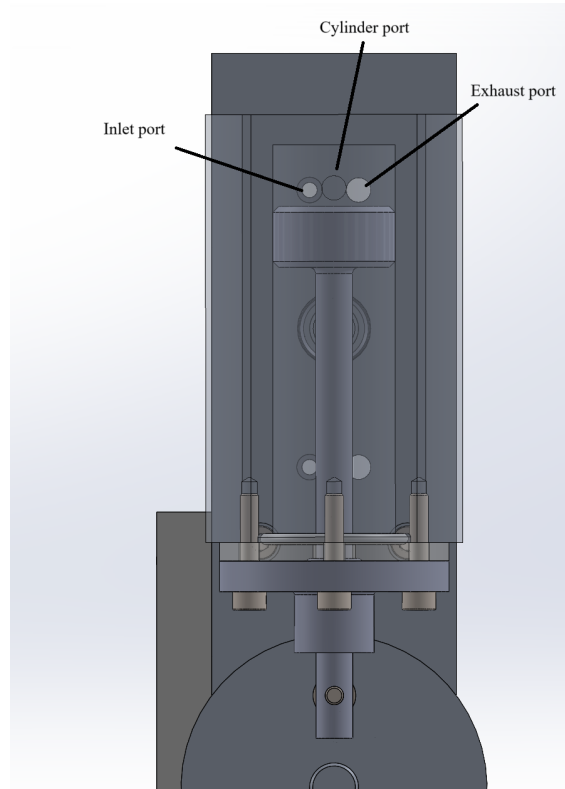


Figure 5: The cylinder and distributor ports

and the cycle repeats.

To increase the consistency of the torque output of the engine, a double-acting design was selected. A double-acting oscillating engine works in the same manner as previously described, but the cycle effectively takes place twice for each crankshaft rotation, with air being allowed to act on both sides of the piston. This is enabled by having a port at either end of the cylinder.

The design raised several engineering problems that needed to be solved. Firstly, airtightness was considered. It was decided that the best way to maintain airtightness between the piston and cylinder wall was to

ensure that both parts were machined precisely. The piston was manufactured first and was then used as a template for boring out the cylinder, ensuring a tight clearance fit that allowed for free movement without unacceptable air leakage. In addition, generous application of lubricating oil during operation would further aid in both objectives. A more difficult problem was sealing the lower half of the cylinder to enable double-acting operation. The solution used was a ‘collar’ around the piston rod with a close-fitting O-ring. This would allow a layer of viscous lithium grease to be maintained around the piston rod, hopefully confining a sufficient air pressure behind the piston head.

Another interface that needed to be airtight was the surface between the flat face of the cylinder and the distributor plate. Several design features were introduced to ensure this. Firstly, both faces were milled flat using a face mill, which produced a very high-quality surface finish with few visible machining marks. This ensured that there would be very close contact between the two faces. Secondly, a spring was included in the design to provide a constant clamping force between the cylinder and the distributor, meaning that no gaps could open during operation. The compression of the spring can be adjusted, allowing the balance between friction and airtightness to be optimised. Lastly, lubricating oil applied to the face would fill any voids between the surfaces that could allow air to escape.

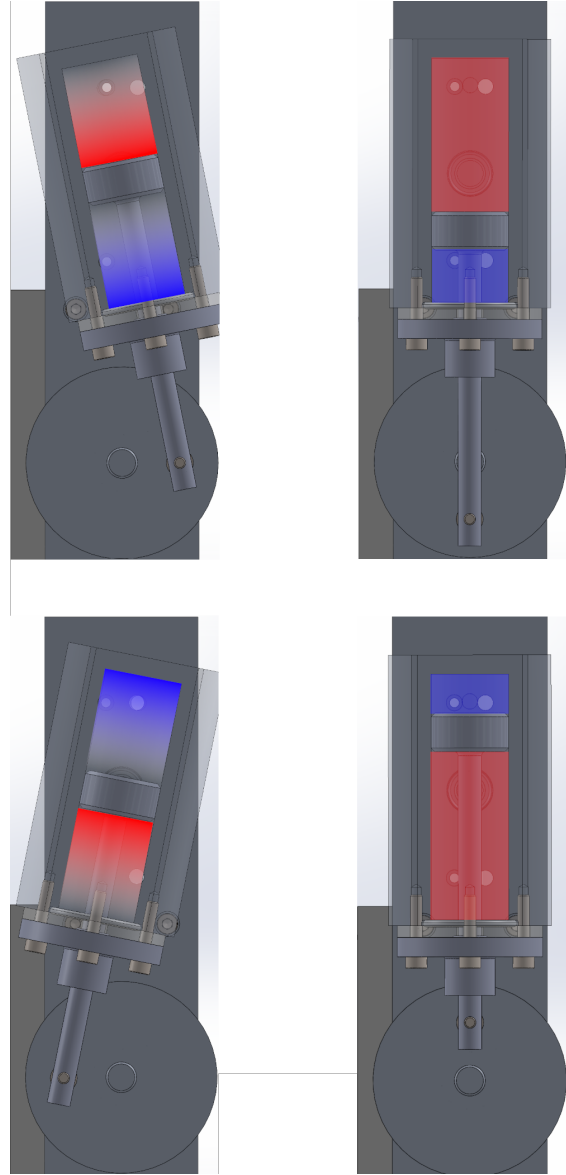


Figure 6: A depiction of a full cycle of the engine

Analysis of other small oscillating engine designs suggested that the engine would run at a relatively high speed. This would likely not provide sufficient torque to accelerate the vehicle from a standstill. With this in mind, the drivetrain had to reduce the engine's speed to a more suitable value. The three main drive designs considered for this application were belt drives, chain drives and gear drives. Belt drives and chain drives require a significant distance between the two wheels and require tensioning. To avoid the need to manufacture the extra components needed for a tensioner, it was decided to use a gear drive. In addition, the compact nature of gears meant that the driveshaft from the engine could be placed directly over the rear axle, allowing the total length of the car to be shortened. This was particularly important considering that the chassis was largely laser-cut, and the largest dimension that could be cut was 600 mm.

To enable the drivetrain to consist of just one stage of simple spur gears, the engine was mounted on its side, with the driveshaft parallel to and above the rear axle. The support structure for the engine was required to securely hold the engine in this orientation without interfering with any moving parts, and additionally house the bearings that support the driveshaft. Based on these requirements, it was decided that 3D printing would be best suited to manufacturing this complex component. For mass production, 3D printing cannot provide sufficiently tight tolerances to ensure that bearings are mounted concentrically, but for a prototype it was decided that the tolerance would suffice, and that the advantages of fast and cheap manufacturing outweighed this disadvantage.

5 Performance Validation

5.1 Gear Selection

Gear selection was performed based on the Lewis equation (Equation 2), and required several iterations to achieve satisfactory results.

$$w = \frac{F \times 10^6}{K_v \times \sigma_p \times \gamma \times m} \quad (2)$$

The pinion is placed on the engine shaft, with the other gear on the rear axle, placing constraints on their physical dimensions. With no accurate way to estimate the rotation speed of the engine, torque was instead used as the basis for the gear ratio.

To calculate the maximum torque produced by the engine, Equation 3 below was used, where P is the maximum pressure of air in the cylinder, A is the area of the piston head, and r is the distance from the piston rod connection to the centre of the crank.

$$T = PAr \quad (3)$$

Using $P = 6$ bar, $A = 3.14 \times 10^{-4} \text{ m}^2$ and $r = 15$ mm, the maximum output torque is 2.83 Nm.

First, plastic gears were considered due to their low mass, but based on the Lewis equation the face width was found to be too high to be practical. Using steel gears, while heavier, would allow for a smaller face width.

After several attempts to find a suitable combination, a set of gears was chosen that met all the requirements. However, upon integration with the CAD assembly, it was found that the gear was too large to fit. An alternative was eventually found, and the final drivetrain parameters are summarised in Table 2. The output torque is estimated to be around 9 Nm assuming no losses in

Table 2: The parameters of the gear drive

Gear	Teeth	PCD (mm)	Minimum Face Width (mm)
RS 521-6402	15	22.5	14.1
RS 182-7847	48	72	9.5

the drivetrain. Gear calculations were performed using a more pessimistic estimate of engine torque that took losses to friction into account. This value, 1.8 Nm was also used for the bearing calculations below.

5.2 Axle Stress Analysis

To ensure that the front and back axles were able to withstand the stress acting on them when attached to the car, calculations were done to find the maximum stresses that the axles experienced, which were compared to the material strength of the axles. First, SolidWorks was used to find where the centre of mass of the vehicle would act, and any unknown forces acting on the axles were found by resolving moments and forces. Next, using the forces calculated previously, a bending moment diagram was plotted for each. The diagrams for the front axle and rear axle are shown in Figures 7 and 8, respectively.

As shown in the figures, the maximum moment experienced by the front axle had a magnitude of 0.25 Nm, and the maximum moment experienced by the rear axle had a magnitude of 0.96 Nm. To find the maximum stress, Equation 4 was used to find

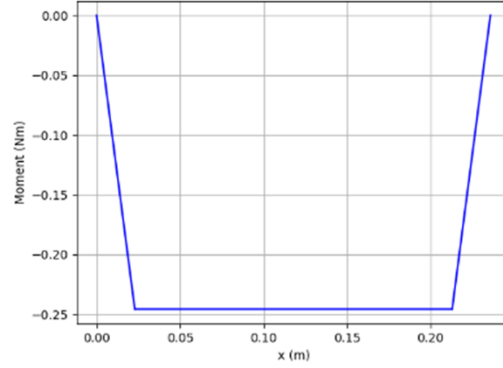


Figure 7: Bending moment diagram for the front axle

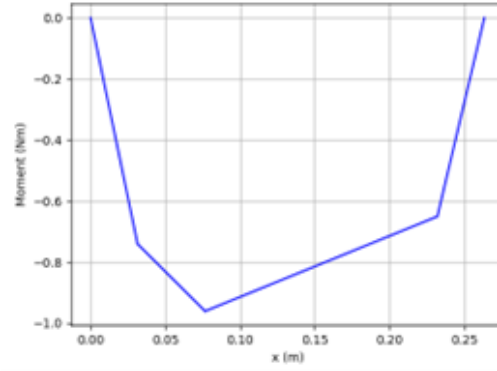


Figure 8: Bending moment diagram for the rear axle

the second moment of area of the axles and Equation 1 was used to calculate the maximum stress experienced.

$$I = \frac{1}{4}\pi r^4 \quad (4)$$

Using these two equations, the maximum stress for the front axle was found to be 1.45 MPa and the maximum stress for the rear axle was 1.22 MPa. Both shafts were manufactured from mild steel bars, which have a material strength of 250 MPa, therefore

Table 3: The safety factors for the axles

Value	Front Axle	Rear Axle
σ_{max} (MPa)	1.45	1.22
SF	172	205

both axles were proven to have a very high strength and durability for use in the car.

Finally, to show the strength of the axles, the safety factors were calculated. These are summarised in Table 3.

5.3 Bearing Selection

To verify that the bearings would last for the 100 hours target lifespan, the dynamic load acting on the bearings was calculated using Equation 5, where C_{min} is the minimum dynamic load rating required from the bearings, F is the force acting on the bearings, and L is the number of revolutions required to meet the target lifespan.

$$C_{min} = FL^{\frac{1}{3}} \quad (5)$$

To use this equation, the rear axle speed was found from the speed of the driveshaft, estimated to be 600 RPM. Dividing this by the gear ratio of 3.2 results in a speed of 187.5 RPM. Then, the CAD model was used to estimate that the mass of the vehicle would be 6 kg. As previously mentioned, the torque from the engine was estimated to be 1.8 Nm, which was used in Equation 6 to calculate the engine's power output.

$$P = T\omega \quad (6)$$

Using the aforementioned values for T and ω results in a power output of 113 W. Dividing

this by the linear speed of the gears, 0.707 ms^{-2} , shows that the force on the bearings from the gears is 160 N. Assuming that total mass of 6 kg is acting on each bearing, a pessimistic overestimate, the total force on each bearing was found to be 170 N.

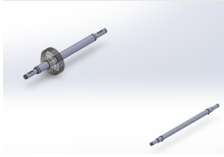
Finally, the life required was found by multiplying the required 100 hours by the rotations per hour of the rear axle. These calculations gave us a final value of 177 N for C_{min} . This allowed us to conclude that the minimum dynamic load required was satisfied by almost any bearing of sufficient size available on RS. We then chose suitable bearings based on their size and mounting type, with confidence that they would not fail before 100 hours of use.

5.4 Flywheel

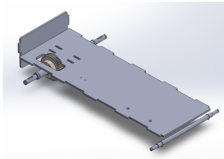
During the initial design stages, it was uncertain whether a flywheel would be needed to maintain engine rotation. It was theorised that the kinetic energy of the drivetrain elements would be sufficient. Modeling these elements as solid cylinders, with $I = \frac{mr^2}{2}$, and assuming a pessimistic engine rotation speed of 2 rads^{-1} , the energy stored in the drivetrain was found to be 5.4 joules.

To model the energy lost as the wheels touch the ground, some assumptions were made: the coefficient of friction μ_s was 0.6 and the time taken for the collision to occur was 0.1 s. Using these assumptions, the energy lost was found to be approximately 2.4 joules. Based on these calculations, it was decided to forgo a flywheel in the design.

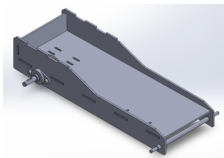
6 Assembly Instructions



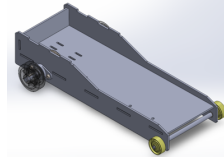
First, mount the 48T gear on the rear axle with a suitable key, and fix it in place with two circlips. Also, install the two 12 mm circlips onto the front axle.



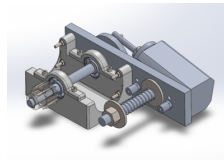
Apply glue to the tabs at the back of the acrylic base and insert into the rear panel.



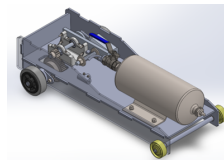
Before the glue dries, apply glue to the remaining tabs and slot on the side panels. Allow the glue to dry before attaching the two bearing units.



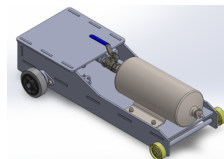
Slide on the rear wheels and secure in place with circlips. Repeat for the front bearing wheels.



Before mounting the engine subassembly, the 15T gear must be mounted on the driveshaft with a key and a circlip.



Attach the engine subassembly, the air tank (with the valve and fittings attached) and the pneumatic T-junction onto the chassis. Connect with 6 mm pneumatic tubing.



Finally, place the lid over the top tabs to conceal the internal components. No glue is used to allow easy access if necessary.

7 Conclusion

This project began with a clear set of requirements outlined in the PDS. These included the development of an air-powered toy car capable of navigating a 25-meter course with specific constraints related to size, weight, and operational performance.

Throughout the ten-week project duration, our team successfully designed, manufactured and assembled the car. Several engine designs were considered, and it was decided that the design that best balanced efficiency, ease of manufacture and design flexibility was an oscillating cylinder engine. The chassis was designed with ease of manufacture in mind, and was therefore laser cut from acrylic. Calculations were performed on each component to ensure that they would perform adequately over the 100-hour design life.

The dimensions of the toy car did not exceed 500 mm in height or 400 mm in width and the CAD model managed weight considerations, minimising it to reduce torque requirements and energy losses. When manufacturing the parts, we produced several turned, milled and laser-cut parts and only three 3D-printed components.

As the design process is now complete, the final stage in the project is to test the car design. The final test will evaluate the real-world performance and functionality of the car and assess the strengths and weaknesses of the design and our manufacturing processes. It will also identify any improvements that could be made.

References

- Engineering Toolbox (2003), ‘Young’s modulus, tensile strength and yield strength values for some materials’, https://www.engineeringtoolbox.com/young-modulus-d_417.html. Accessed December 2023.
- Mohammed, I. K. (2023), *DMF2 Project Brief*, Imperial College London.

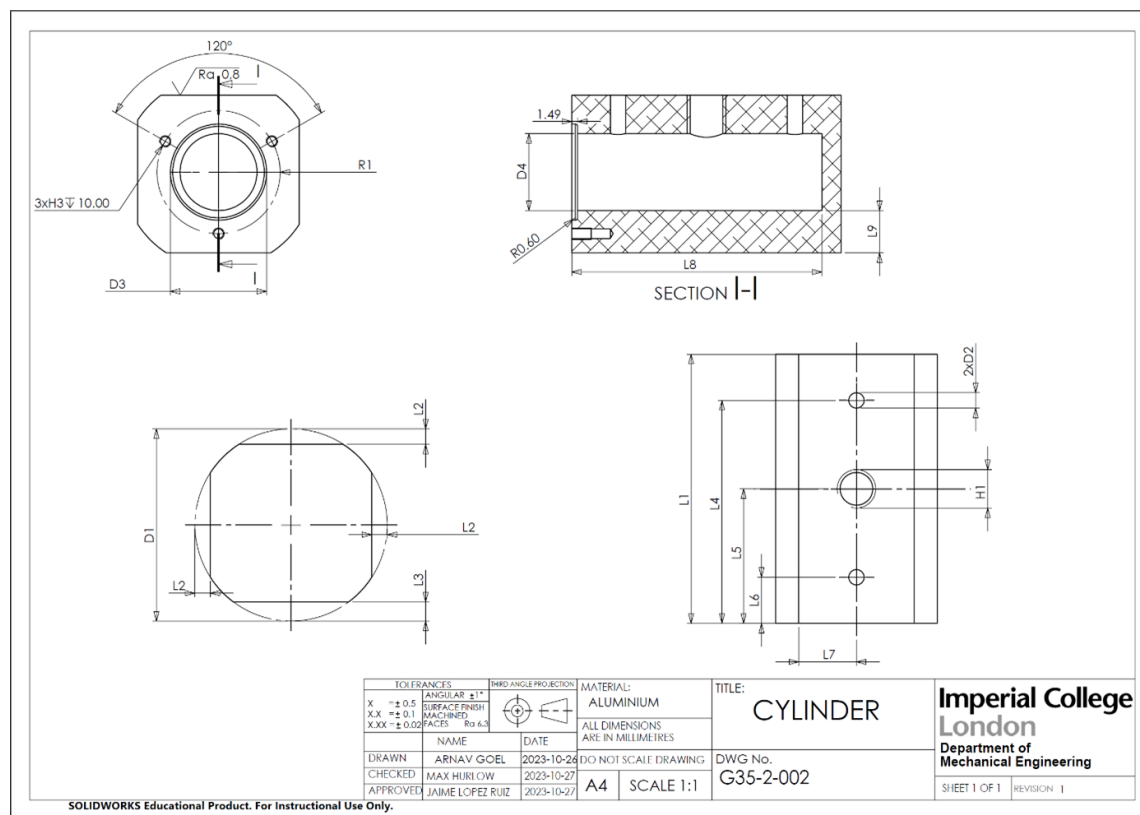
Appendices

A Measurement Report

This appendix details the measurement report for the cylinder and distributor. It should be noted that some details, like surface roughness and the depths of holes, could not be measured due to the limitations of the measuring equipment available.

The tables below reference dimensions on the included part drawings.

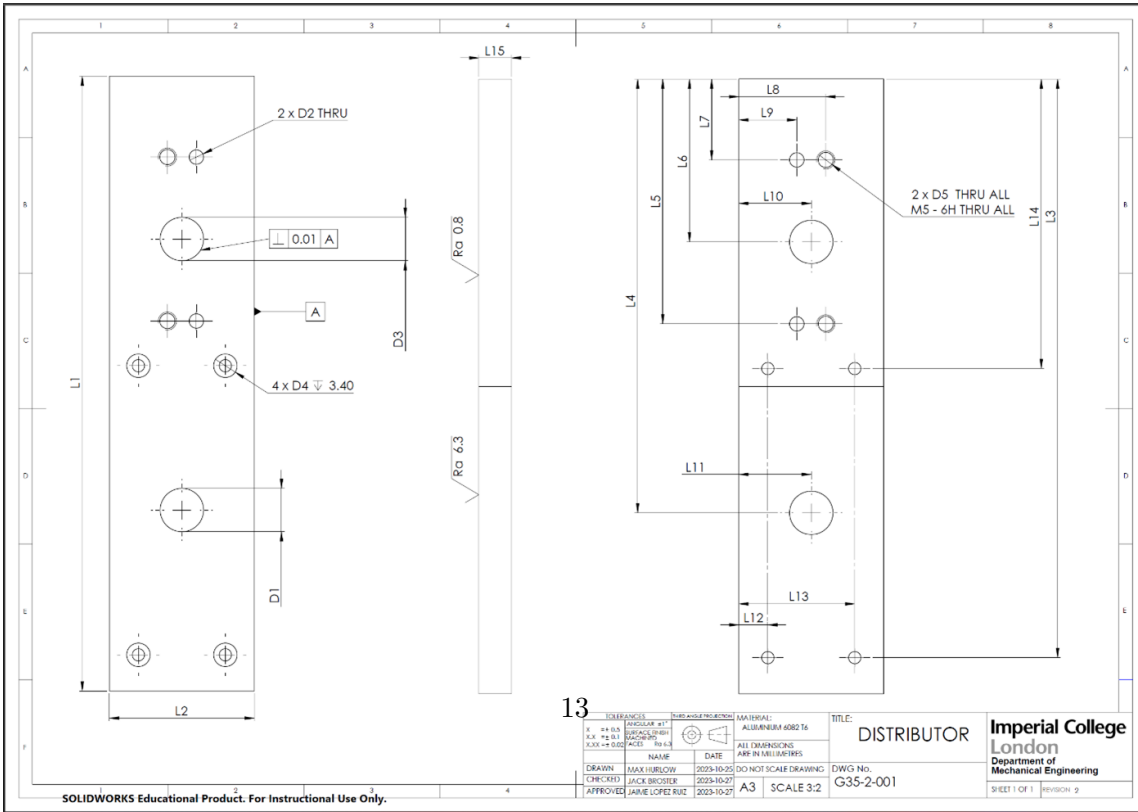
A.1 Cylinder



Measurement summary for the cylinder

Dimension ID	Nominal Value (mm)	Measuring Tool	Actual Measurement (mm)	Comments
L1	70 ± 0.1	Vernier Callipers	76.28 ± 0.01	Outside tolerance (will be fixed in future)
L2	4 ± 0.5		3.98 ± 0.01	Within tolerance
L3	5 ± 0.02		5.05 ± 0.01	Outside tolerance (still acceptable)
L4	58 ± 0.02		59.95 ± 0.01	Within tolerance
L5	35 ± 0.02		37.12 ± 0.01	
L6	12 ± 0.02		13.91 ± 0.01	
L7	15 ± 0.02		14.92 ± 0.01	
L8	65 ± 0.5		64.99 ± 0.01	
L9	11 ± 0.02		10.98 ± 0.02	
D1	50 ± 0.5		50.05 ± 0.01	
D2	4 ± 0.1		3.99 ± 0.01	
D3	25 ± 0.1		24.95 ± 0.01	
D4	20 ± 0.02		19.98 ± 0.01	
R1	16 ± 0.02		16 ± 0.01	

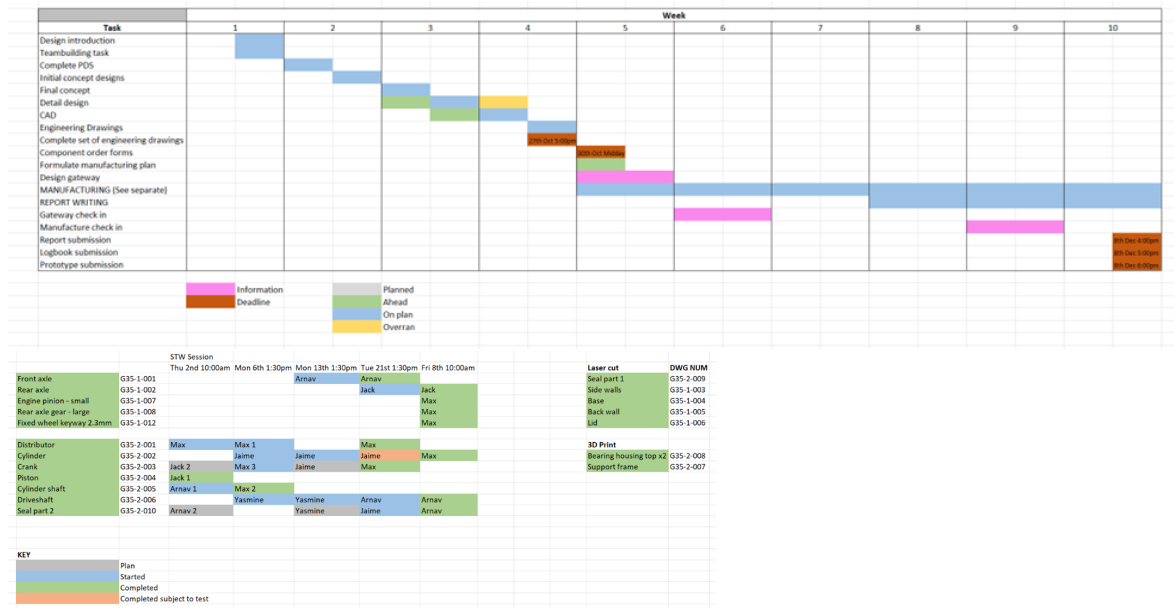
A.2 Distributor



Measurement summary for the distributor

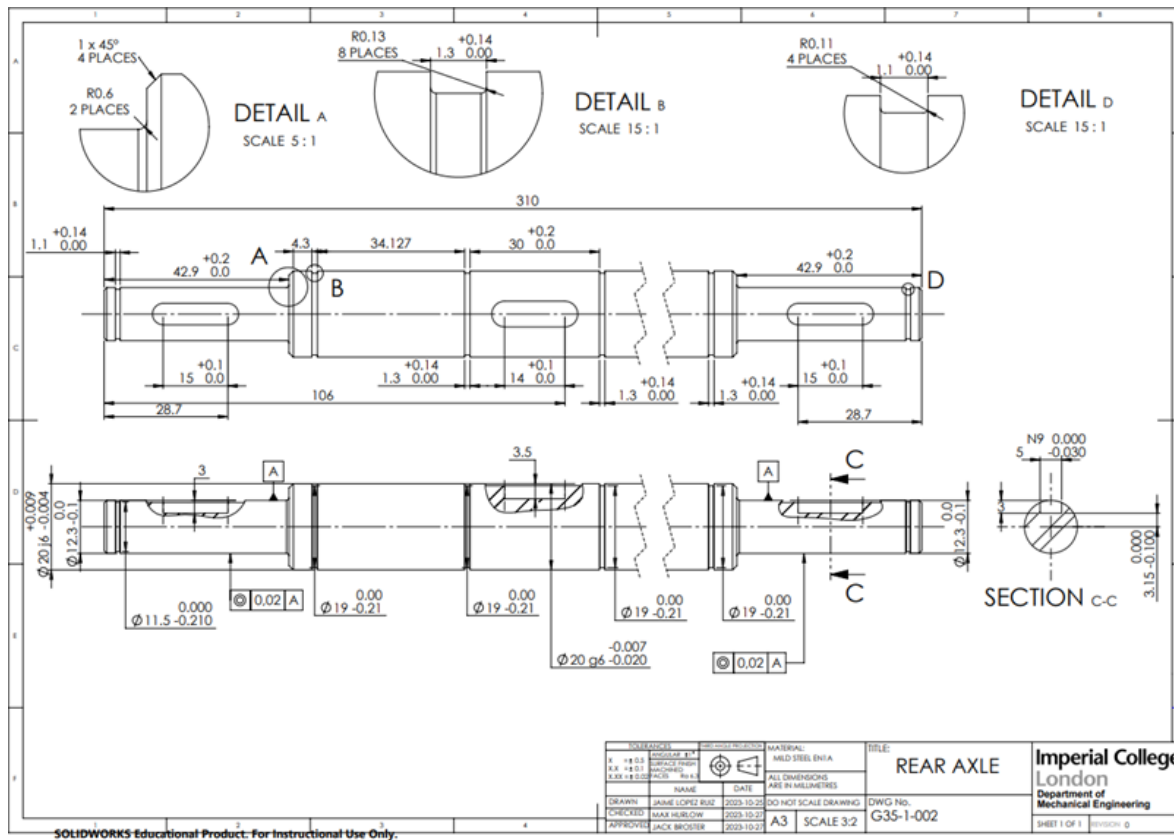
Dimension ID	Nominal Value (mm)	Measuring Tool	Actual Measure- ment (mm)	Comments
L1	170 ± 0.5	Metre rule	170 ± 0.5	Within tolerance
L2	40 ± 0.5	Vernier callipers	39.91 ± 0.01	
L3	160 ± 0.1	Metre rule	160 ± 0.5	
L4	120 ± 0.1	Vernier callipers	124.7 ± 0.01	Outside of tolerance (still acceptable)
L5	67.65 ± 0.02		67 ± 0.01	
L6	45 ± 0.02		45 ± 0.01	Within tolerance
L7	22.35 ± 0.02		22.35 ± 0.01	
L8	24 ± 0.02		23.95 ± 0.01	Outside of tolerance (still acceptable)
L9	16 ± 0.02		16.03 ± 0.01	
L10	20 ± 0.02		20 ± 0.01	Within tolerance
L11	20 ± 0.02		20 ± 0.01	
L12	8 ± 0.02		8.02 ± 0.01	
L13	32 ± 0.02		32 ± 0.01	
L14	80 ± 0.1		80.15 ± 0.01	Outside of tolerance (still acceptable)
L15	9 ± 0.5		8.85 ± 0.01	Within tolerance
D1	$12^{+0.1}_{-0.0}$		11.84 ± 0.01	Outside of tolerance (still acceptable)
D2	4 ± 0.02		3.79 ± 0.01	
D3	12 H7		11.86 ± 0.01	
D4	6.5 ± 0.02		6.81 ± 0.01	
D5	9 ± 0.02		8.99 ± 0.01	Within tolerance

B Project Timeline

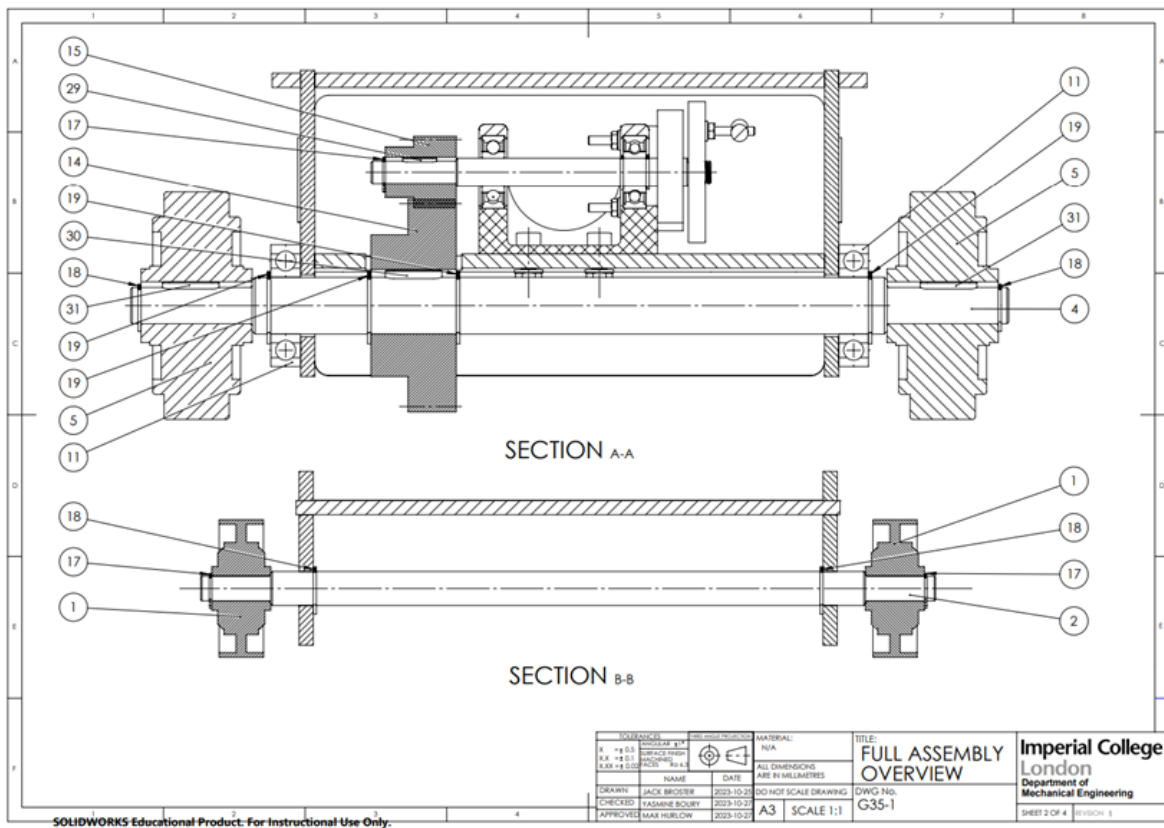
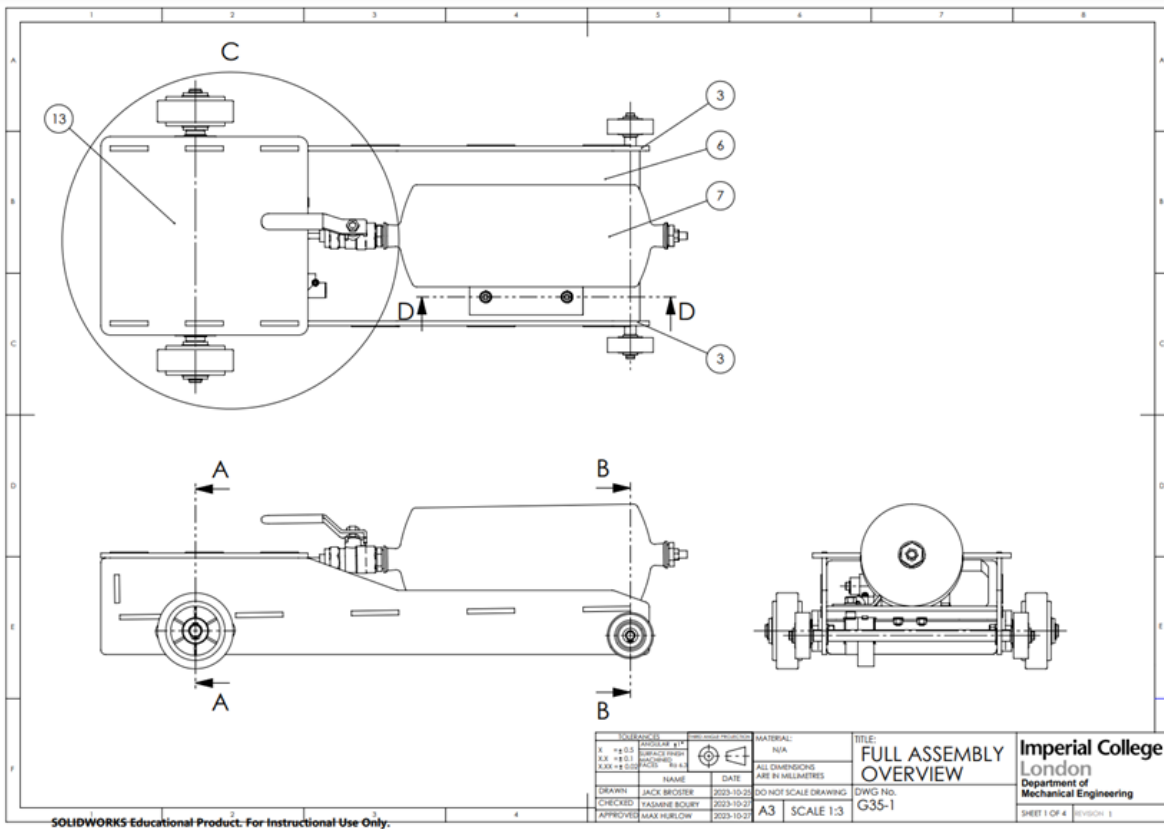


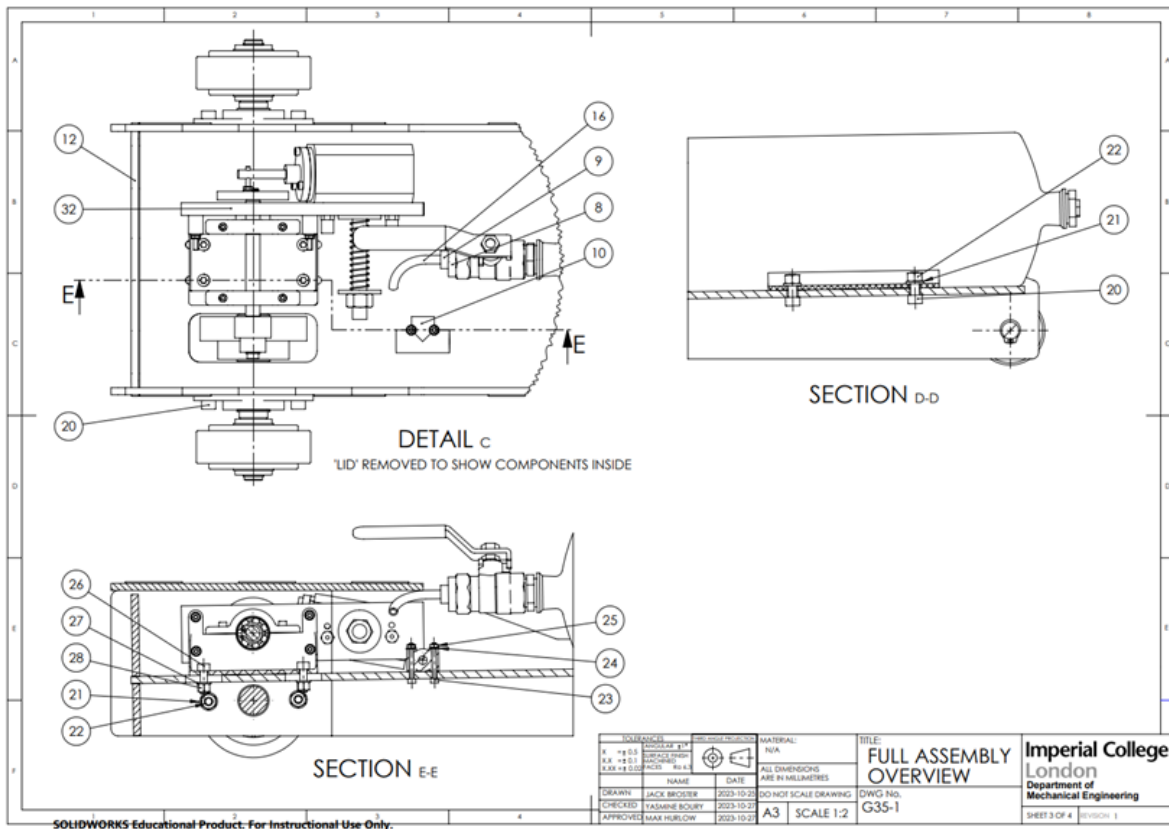
C Engineering Drawings

C.1 Rear Axle



C.2 General Assembly





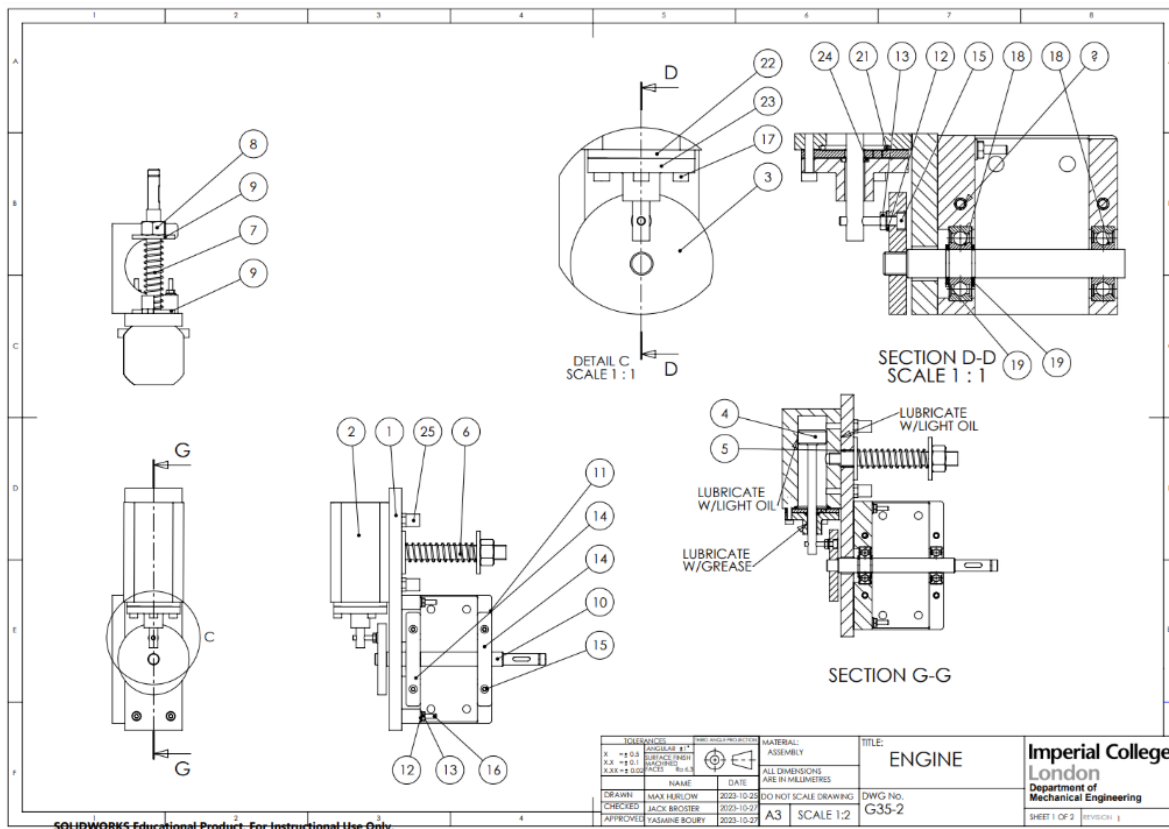
ITEM NO.	PRTRNUM	DESCRIPTION	QTY.
1	G35-1-011	FRONT BEARING WHEEL	2
2	G35-1-001	FRONT AXLE	1
3	G35-1-003	SIDE WALL	2
4	G35-1-002	REAR AXLE	1
5	G35-1-012	REAR FIXED WHEEL	2
6	G35-1-004	ACRYLIC BASE	1
7	G35-1-010	AIR RESERVOIR	1
8	G35-1-013	REDUCING CONNECTOR 0.5 - 0.125 INCH (RS-220-3350)	1
9	G35-1-014	REDUCING CONNECTOR 0.125 INCH - 6MM (RS-299-3819)	1
10	G35-1-016	T-JUNCTION 6MM (RS-916-0909)	1
11	G35-1-009	REAR AXLE BEARING - SFT20-4N-D (RS-109-949)	2
12	G35-1-005	BACK WALL	1
13	G35-1-006	LID	1
14	G35-1-008	REAR AXLE GEAR - 48T (RS-182-7847)	1
15	G35-1-007	REAR AXLE PINION - 15T (RS-521-6402)	1
16	G35-1-015	PNEUMATIC TUBING 6MM (RS-483-4970)	1

ITEM NO.	PRTRNUM	DESCRIPTION	QTY.
17	G35-1-017	8MM EXTERNAL CIRCLIP (RS-289-225)	3
18	G35-1-018	12MM EXTERNAL CIRCLIP	4
19	G35-1-019	20MM EXTERNAL CIRCLIP	4
20	G35-1-026	M6x16 SOCKET HEAD CAP SCREW	6
21	G35-1-027	M6 PLAIN WASHER	6
22	G35-1-028	M6 HEX FULL NUT	6
23	G35-1-020	M3x25 SOCKET HEAD CAP SCREW	2
24	G35-1-021	M3 PLAIN WASHER	2
25	G35-1-022	M3 HEX FULL NUT	2
26	G35-1-023	M5x16 SOCKET HEAD CAP SCREW	4
27	G35-1-024	M5 PLAIN WASHER	4
28	G35-1-025	M5 HEX FULL NUT	4
29	G35-1-031	12x3x3 PINION GEAR KEY (RS-302-3949)	1
30	G35-1-029	20x6x6 SHAFT GEAR KEY (RS-302-4144)	1
31	G35-1-030	20x5x5 REAR WHEEL KEY (RS-302-4087)	2
32	G35-2	ENGINE SUBASSEMBLY	1

TOLERANCES		DIMENSIONS		MATERIAL		TITLE	
± 0.5	± 0.5	± 0.5	± 0.5	N/A		BILL OF MATERIALS	Imperial College London
± 0.5	± 0.5	± 0.5	± 0.5	ALL DIMENSIONS ARE IN MILLIMETRES		DO NOT SCALE DRAWING	Department of Mechanical Engineering
DRAWN: JACK BOWLER		DATE: 2023-10-25		DWG No: G35-1		SHEET 4 OF 4	
CHECKED: FADJANE BOURY		DATE: 2023-10-25		SCALE 1:3		REVISION: 1	
APPROVED: MAX HURLLOW		DATE: 2023-10-25					

SOLIDWORKS Educational Product. For Instructional Use Only.

C.3 Engine Subassembly



Further part drawings are included in the measurement report.